

Low Level Design

HR ANALYTICS

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1. Introduction

1.1 What is Low-Level design document?

The Low-Level Design (LLD) document is a detailed blueprint that outlines the specific components, interfaces, and processes of a system. It provides a granular view of how each module within a project functions and interacts with other components. This document serves as a guide for developers and data scientists, detailing how the architecture will be implemented and how the project will function on a technical level.

1.2 Scope

The scope of this LLD document is to define the components and interactions within the HR Analytics project aimed at predicting employee attrition. It covers:

- The detailed architecture of the system.
- Descriptions of each component and its functionality.
- Data flow between modules.
- Unit test cases for ensuring the correctness and reliability of the components.

The goal is to provide sufficient detail to guide developers in implementing the system while ensuring that the overall project objectives are met.

2. Architecture

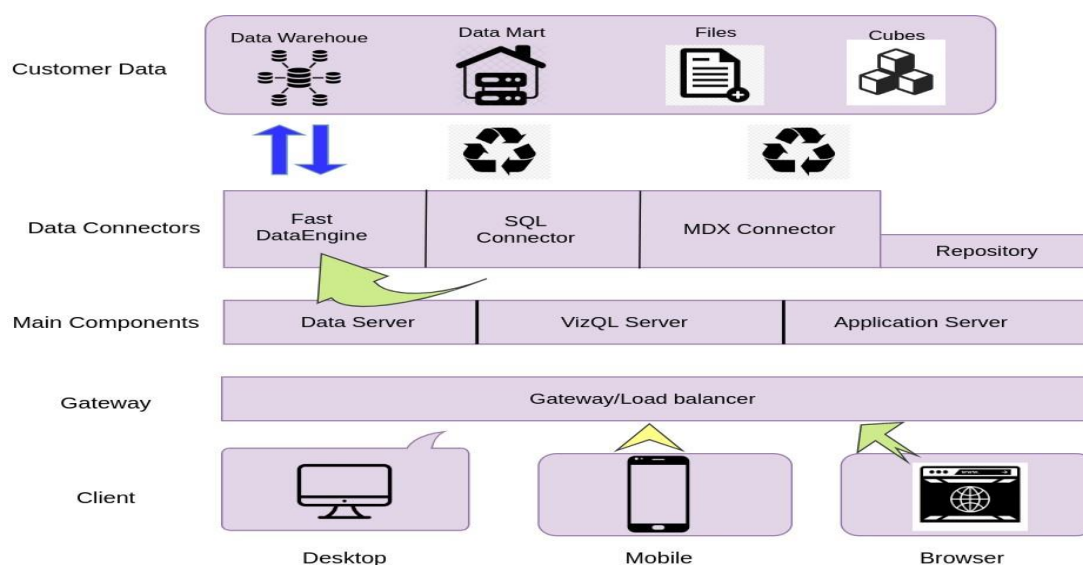
The architecture of the HR Analytics project is designed to facilitate the systematic analysis of employee data to predict attrition. It is composed of several key modules, each responsible for specific tasks in the data processing pipeline. The architecture can be visualized as a series of interconnected components that work collaboratively to deliver actionable insights.

2.1 Overview of the Architecture

The architecture follows a modular design, allowing for flexibility, scalability, and maintainability. The main components include:

- **Data Collection Module**
- **Data Preprocessing Module**
- **Exploratory Data Analysis (EDA) Module**
- **Feature Engineering Module**
- **Model Development Module**
- **Model Evaluation Module**
- **Reporting Module**

Each of these components plays a vital role in transforming raw employee data into valuable insights for HR decision-making.



3. Architecture Description

3.1. Data Description

1. Data Collection Module

- **Purpose:** To gather and store employee data from various sources.
- **Functionality:** Collects historical employee records, which may include attributes like demographics, job satisfaction, compensation, performance ratings, and tenure. It can interface with databases or external data sources (e.g., CSV files, HR management systems) to retrieve the necessary data.
- **Data Flow:**
 - Input: Raw employee data is ingested from various sources.
 - Output: The collected data is stored in a structured format for preprocessing.

2. Data Preprocessing Module

- **Purpose:** To prepare the raw data for analysis by cleaning and transforming it.
- **Functionality:** Handles missing values through techniques such as imputation or removal, encodes categorical variables into a numerical format, normalizes or scales continuous variables for consistency, and identifies and manages outliers.
- **Data Flow:**
 - Input: Raw dataset from the data collection module.
 - Output: A cleaned and transformed dataset ready for exploratory analysis.

3. Exploratory Data Analysis (EDA) Module

- **Purpose:** To provide insights into the data through statistical analysis and visualizations.
- **Functionality:** Computes descriptive statistics to summarize the dataset and generates visualizations (e.g., histograms, scatter plots) to identify trends and

relationships among variables. Performs correlation analysis to examine the strength and direction of relationships between variables.

- **Data Flow:**
 - Input: Cleaned dataset from the data preprocessing module.
 - Output: EDA reports and visualizations that inform feature selection and model development.

4. Feature Engineering Module

- **Purpose:** To select and create relevant features for the predictive model.
- **Functionality:** Analyzes the results of the EDA to select features that significantly impact employee attrition, using techniques such as ANOVA (Analysis of Variance) and Chi-Square tests. Creates new features if necessary (e.g., categorizing employees into salary bands or calculating tenure).
- **Data Flow:**
 - Input: Insights from the EDA module and the cleaned dataset.
 - Output: A refined set of features prepared for model training.

5. Model Development Module

- **Purpose:** To train machine learning models to predict attrition.
- **Functionality:** Implements various machine learning algorithms (Logistic Regression, Random Forest, XGBoost) to build predictive models based on the selected features and utilizes cross-validation techniques to ensure models generalize well to unseen data.
- **Data Flow:**
 - Input: Selected features and target variable (attrition) from the feature engineering module.
 - Output: Trained machine learning models ready for evaluation.

6. Model Evaluation Module

- **Purpose:** To evaluate the performance of the trained models and identify the best-performing one.
- **Functionality:** Assesses models using various metrics (accuracy, precision, recall, and ROC-AUC score) to quantify how well they predict employee attrition and compares the performance of multiple models to determine which one provides the best predictive capabilities.
- **Data Flow:**
 - Input: Trained models and a test dataset.
 - Output: Evaluation reports summarizing model performance and recommendations for the best model.

7. Reporting Module

- **Purpose:** To generate comprehensive reports that summarize the findings from the analysis.
- **Functionality:** Compiles insights from the EDA, model evaluations, and feature importance to create a detailed report for HR stakeholders. Includes visualizations and key metrics to help decision-makers understand the factors influencing attrition.
- **Data Flow:**
 - Input: Insights and metrics from the EDA and model evaluation modules.
 - Output: Final report document ready for presentation to stakeholders.

3.2 Deployment Module

Purpose : The deployment module focuses on making the system and its components accessible to users, ensuring the insights gained from the analysis can be effectively utilized by HR professionals.

Functionality :

- **Environment Setup:** The deployment involves configuring the necessary environment where the Python and Jupyter Notebook code can run smoothly, ensuring that all dependencies and libraries are available.
- **Model Deployment:** The trained machine learning models are saved and can be loaded for predictions on new data, allowing HR teams to input employee attributes and receive attrition predictions.
- **Output Management:** The results of the analysis, including visualizations and reports, are prepared for sharing with stakeholders. This can be done through exporting reports in HTML or PDF formats or through interactive dashboards if implemented in the future.
- **Version Control:** The project is maintained using a version control system (e.g., GitHub) to track changes, collaborate effectively, and ensure all team members have access to the latest code and documentation.

Data Flow :

- **Input:** All components of the architecture, including trained models, cleaned data, EDA results, and reporting outputs.
- **Output:** A deployed system that provides HR professionals with predictions and insights into employee attrition.

4. Unit Test Cases

Unit tests are critical for ensuring the correctness of each module within the HR Analytics project.

4.1 Data Preprocessing Module Tests

- **Test Case 1:** Test for handling missing values.
 - **Input:** A dataset with missing values in various columns.
 - **Expected Output:** The output dataset should have no missing values.
- **Test Case 2:** Test for correct encoding of categorical variables.
 - **Input:** A dataset with categorical variables (e.g., job role).
 - **Expected Output:** Categorical variables should be converted to numeric format correctly.

4.2 EDA Module Tests

- **Test Case 3:** Test for generating summary statistics.
 - **Input:** Cleaned dataset.
 - **Expected Output:** The output should include descriptive statistics such as mean, median, and standard deviation for numerical columns.
- **Test Case 4:** Test for correlation analysis.
 - **Input:** A dataset with several features.
 - **Expected Output:** A correlation matrix should be produced, showing the relationships between features.

4.3 Model Development Module Tests

- **Test Case 5:** Test for model training.
 - **Input:** A feature set and corresponding labels for attrition.
 - **Expected Output:** A trained model object should be created without errors.
- **Test Case 6:** Test for model predictions.
 - **Input:** A set of features for a new employee.
 - **Expected Output:** The model should return a predicted attrition value (e.g., 0 or 1).

4.4 Reporting Module Tests

- **Test Case 7:** Test for report generation.
 - **Input:** Model evaluation metrics and EDA results.
 - **Expected Output:** A report document should be generated containing all key findings and insights.